to select candidate chunks for memory updating:

$$a_{t,i} \sim \pi_\phi^{\text{sel}}(\cdot \mid q, c_i, m_t), \quad a_{t,i} \in \{\textsc{Skip}, \textsc{Update}\}. \tag{2}$$

The chunks predicted as UPDATE form the candidate set

$$\mathcal{C}_t = \{c_i \in \mathcal{W}_t \mid a_{t,i} = \textsc{Update}\}, \tag{3}$$

which is forwarded to System 2.

3.2.2 System 2: Speculative Verification and Memory Consolidation

Inspired by speculative decoding's acceptance rule (Leviathan et al., 2023; Chen et al., 2023a), System 2's binary policy π_θ^{ver} assesses whether selected chunks warrant memory updates. For candidate indices $\mathcal{I}_t = \{i \mid c_i \in \mathcal{C}_t\}$, the action-level acceptance probability is

$$\alpha_{t,i} = \min\left(1, \frac{\pi_\theta^{\text{ver}}(\textsc{Update} \mid q, c_i, m_t)}{\pi_\phi^{\text{sel}}(\textsc{Update} \mid q, c_i, m_t)}\right). \tag{4}$$

Independently draw $\xi_{t,i} \sim \mathcal{U}(0,1)$ for each candidate and accept it if $\xi_{t,i} \le \alpha_{t,i}$.

If any candidate is accepted, commit the first accepted chunk in document order and generate its replacement memory:

$$m_{t+1} \sim \pi_\theta^{\text{mem}}(\cdot \mid q, c_{i_t^\star}, m_t), \tag{5}$$

where $i_t^\star = \min\{i \in \mathcal{I}_t \mid \xi_{t,i} \le \alpha_{t,i}\}$. As a special case, if no candidate is accepted, retain the current memory and advance past the window. Otherwise, discard remaining predictions conditioned on m_t and resume immediately after the committed chunk with m_{t+1}. Answering uses the query and final memory; Algorithm 3 in Appendix B.4 gives the complete transitions.

3.3 Training of MemDuet

Figure 3 summarizes three-stage training. Stage I trains memory-conditioned evidence selection. Stage II freezes System 1 and trains System 2 to verify candidates, rewrite memory, and generate answers. Stage III jointly optimizes both systems with task rewards, coordinating selection with verification and memory updates.

3.3.1 Stage I: System 1 Supervised Warm-up

We initialize System 1 on HotpotQA (Yang et al., 2018) by aligning gold supporting facts with document chunks. Evidence-containing chunks are labeled UPDATE, and other chunks SKIP.

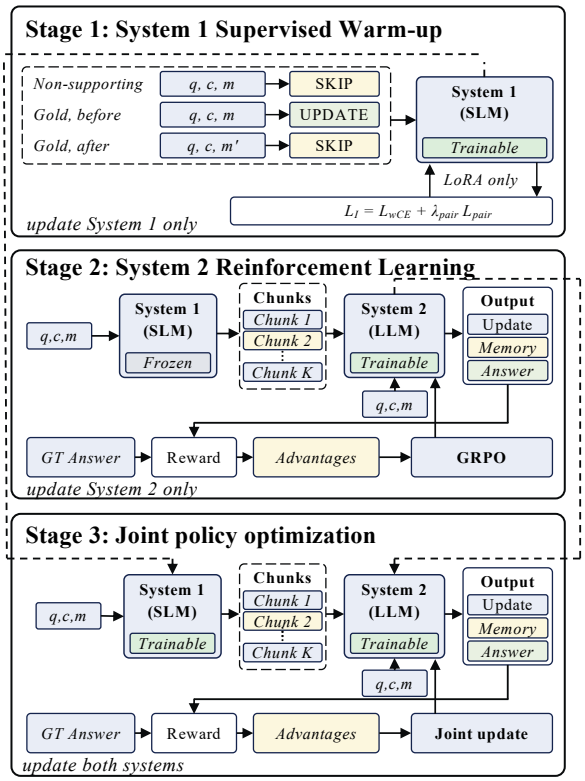


Figure 3: Three-stage training of MemDuet. Stage I trains System 1 with weighted classification and a paired-memory hinge loss. Stage II freezes System 1 and optimizes System 2 with GRPO. Stage III jointly optimizes both systems using task rewards.

For each UPDATE example (q, c, m), let m' be the memory after incorporating the evidence from c. We retain the original example and add a post-update negative:

$$(q, c, m) \longrightarrow \textsc{Update}, \quad (q, c, m') \longrightarrow \textsc{Skip}. \tag{6}$$

We train on the original examples and post-update negatives using the memory-contrastive objective:

$$\mathcal{L}_{\text{Stage I}}(\phi) = \mathcal{L}_{\text{wCE}} + \lambda_{\text{pair}}\mathcal{L}_{\text{pair}}, \quad \mathcal{L}_{\text{pair}} = \mathbb{E}_{\mathcal{P}}\left[\max(0, \gamma - \Delta s_\phi)\right]. \tag{7}$$

Here, $\mathcal{L}_{\text{wCE}}$ is class-weighted action cross-entropy; $\mathcal{P}$ contains matched memory pairs, with $\Delta s_\phi = s_\phi(q, c, m) - s_\phi(q, c, m')$. The score s_ϕ is the UPDATE-minus-SKIP logit difference, and γ is the margin. Hyperparameters are reported in Appendix B.1.

3.3.2 Stage II: System 2 Reinforcement Learning

We freeze System 1 at ϕ_{SFT} and optimize all System 2 parameters θ by reinforcement learning over